

Semi-supervised Modelling

UBCO MDS — DATA 573



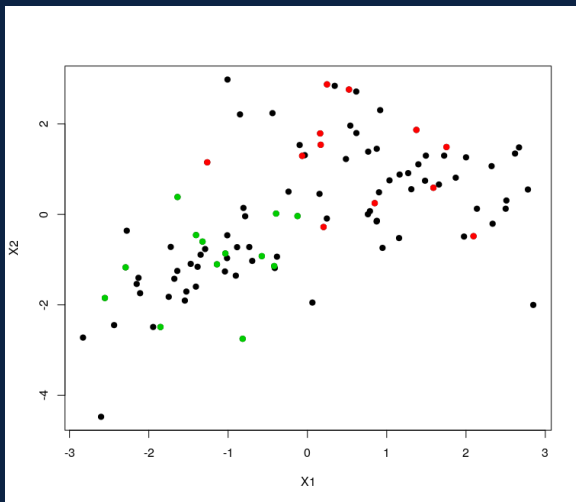


- ▶ We have focussed on supervised methods from 570 through 572. This includes taking known (labeled) responses and fitting a model — you might then predict for some unknown (unlabeled) responses using that fitted model.
- ▶ The bulk of 573 has been focussed on unsupervised methods. This includes taking unlabeled responses and fitting a model.
- ▶ Let's motivate an alternative through some examples...

Supervised vs Unsupervised



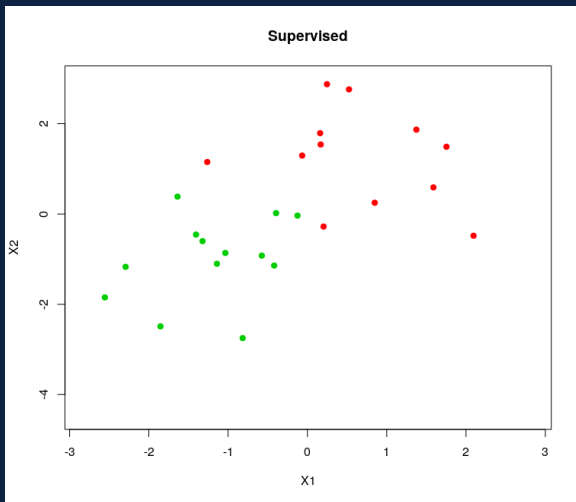
- Suppose that 25% of cases have labeled response from the following data



Supervised estimation



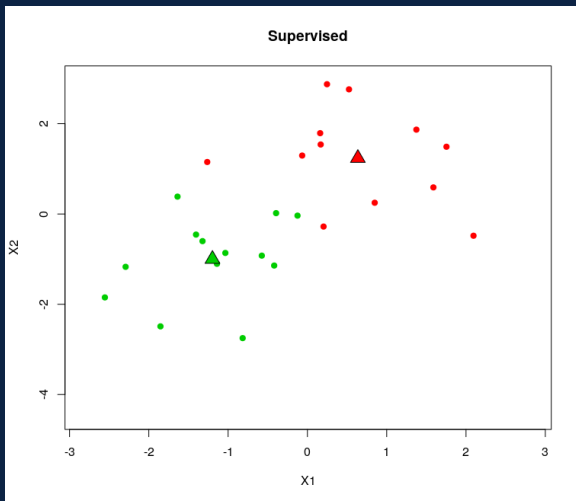
- ▶ Let's calculate the means from each group in a supervised manner



Supervised estimation



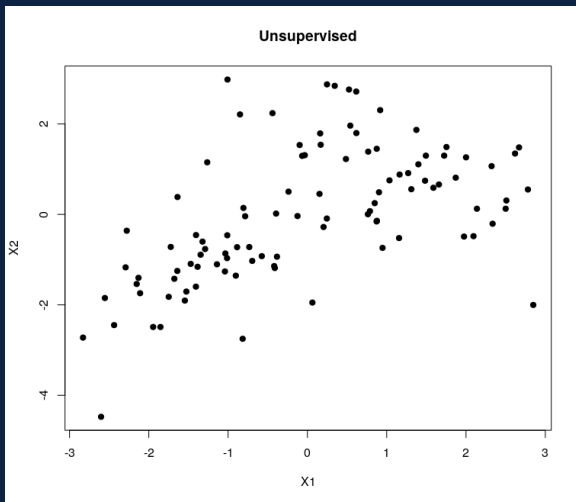
- ▶ Let's calculate the means from each group in a supervised manner



Unsupervised estimation



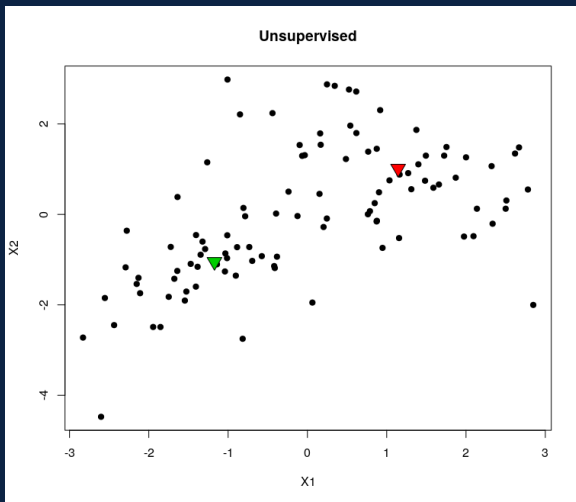
- Let's calculate the means for estimated groups in an unsupervised manner (using mclust)



Unsupervised estimation



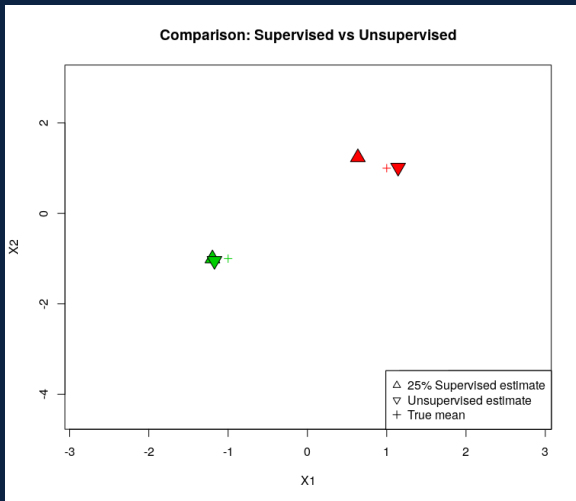
- Let's calculate the means for estimated groups in an unsupervised manner (using mclust)



Supervised vs Unsupervised



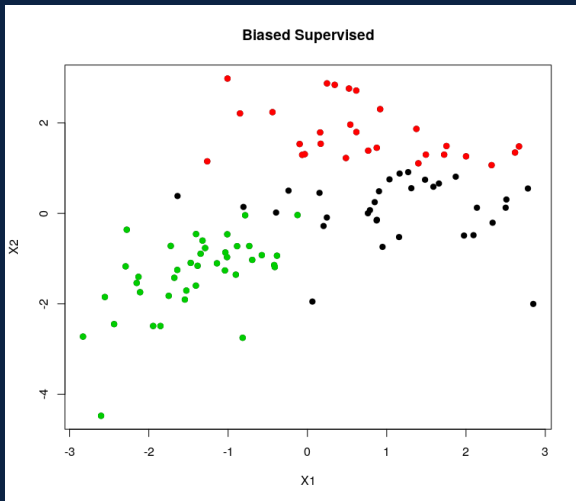
- Here are the group means on the same plot



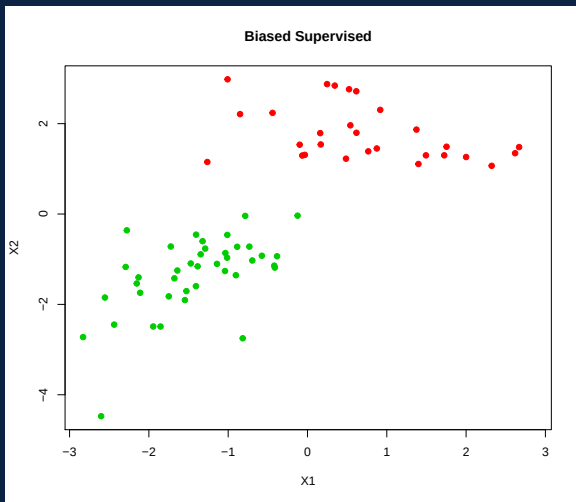


- ▶ And that was with 25% randomly selected to have known labels.
- ▶ Suppose instead that your labeling process was biased...
- ▶ One natural bias that can happen in realistic scenarios is that clear cases are labeled, and less clear cases might be left for an algorithm to sort out.

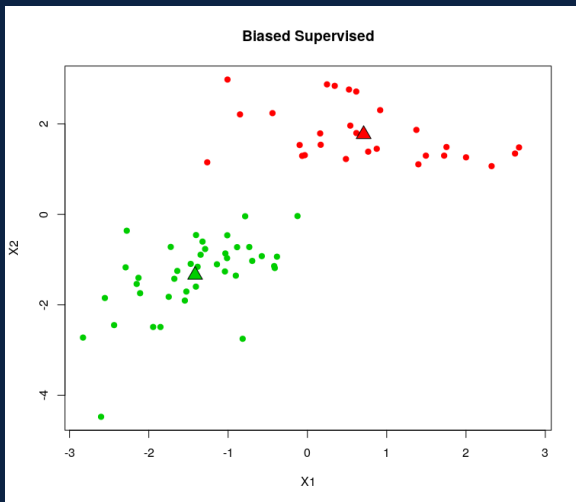
- Suppose that the following cases have been hand-labeled



- We then estimate the means...



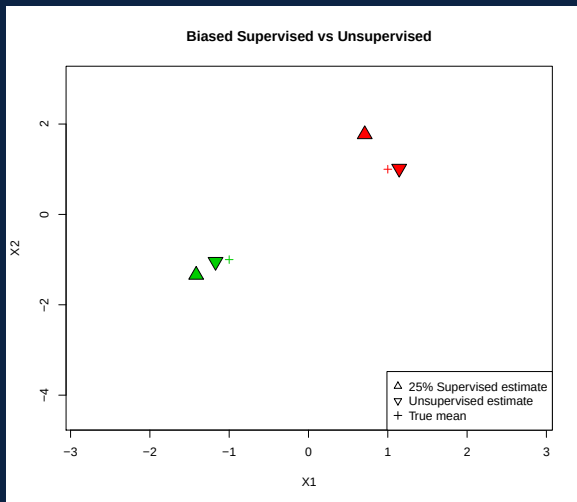
- We then estimate the means...



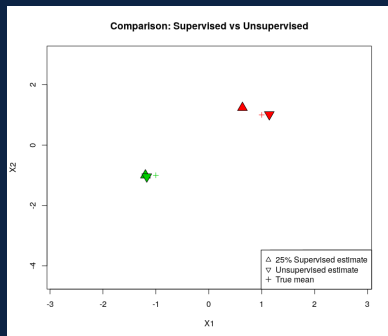
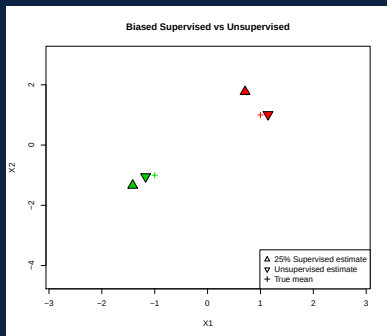
Bias Supervised vs Unsupervised



- And again compare to unsupervised...



Side by side



- ▶ Of course, if the group structure is less clear, unsupervised methods can fail horribly on this type of estimation example.
- ▶ In which case supervised methods will outperform.
- ▶ Furthermore, it seems like there should be **some** way to take advantage of labeled responses to improve on (blind) unsupervised methods.



- ▶ There is a third option....or rather, a third set of options¹.
- ▶ Semi-supervised methods use **both** labeled and unlabeled observations in tandem during the model fitting process.
- ▶ While such an approach is usable in several classification models, we will focus on mixture models (surprise, surprise).

¹Vrbik, I., & McNicholas, P. D. (2015). Fractionally-supervised classification. *Journal of Classification*, 32(3), 359-381.

- ▶ Suppose X_u contains all the predictors with unlabeled response (unobserved y_u), and X_l contains all predictors with labeled response (observed y_l).

Paradigm	Avail. Data	Estimate $\hat{f}$ with	Provides
Supervised	y_l, X_l, X_u	y_l, X_l	$\hat{y}_l, \hat{y}_u$
Unsupervised	y_l, X_l, X_u	X_l, X_u	$\hat{y}_l, \hat{y}_u$
Semi-supervised	y_l, X_l, X_u	y_l, X_l, X_u	$\hat{y}_l, \hat{y}_u$

- ▶ For mixture models:
 - ▶ Supervised = Discriminant Analysis
(Gaussian unconstrained = QDA)
 - ▶ Unsupervised = Model-based clustering
(Gaussian unconstrained = Mclust 'VVV' model)
 - ▶ Semi-supervised = sometimes referred to as Model-based classification



- ▶ One reason to focus on mixture models (beyond my affinity for them), is that implementing semi-supervised modelling is relatively trivial.
- ▶ The basic idea is to take your labeled cases (y_l) and both pre-determine group membership, as well as not allow that group membership to change during the model-fitting process.
- ▶ This is easy to implement in the EM algorithm...let's quickly review

Recall: EM Algorithm for Clustering



1. Start the algorithm with random values for $\hat{z}_{ig}$. (there are alternative starting options)
2. Assuming those $\hat{z}$ are correct, estimate parameters μ_g and σ_g (via MLEs — hence, this is the **maximization** of EM)
3. Assuming those parameters are correct, find the expected value of group memberships

$$\hat{z}_{ig} = \frac{\pi_g \phi(\mathbf{x}_i \mid \mu_g, \sigma_g)}{\sum_{g=1}^G \pi_g \phi(\mathbf{x}_i \mid \mu_g, \sigma_g)}$$

(this is the **expectation** of EM)

4. Repeat 2. and 3. until 'changes' are minimal. (The log-likelihood of the model is monitored for convergence)

EM Algorithm for Semi-supervised Classification



1. Start the algorithm with random values for $\hat{z}_{ig}$ where $i \in X_u$.
For $j \in X_l$, set $\hat{z}_{jg} = 1$ if $y_l = g$, otherwise 0
2. Assuming those $\hat{z}_{ig}$ are correct, estimate parameters μ_g and σ_g
3. Assuming those parameters are correct, find the expected value of group memberships for $i \in X_u$

$$\hat{z}_{ig} = \frac{\pi_g \phi(\mathbf{x}_i \mid \mu_g, \sigma_g)}{\sum_{g=1}^G \pi_g \phi(\mathbf{x}_i \mid \mu_g, \sigma_g)}$$

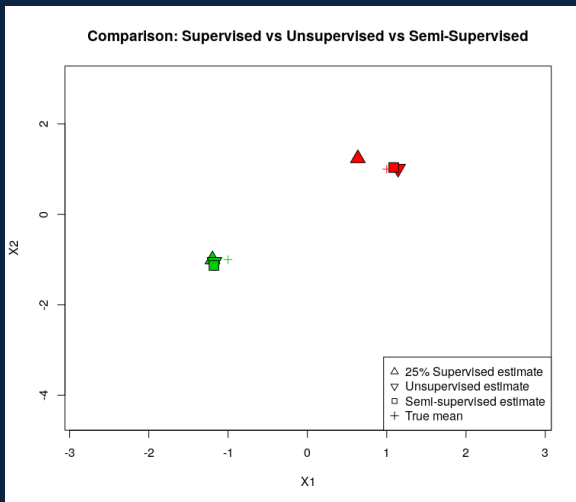
(aka, update unlabeled estimates, leave labeled estimates as initialized)

4. Repeat 2. and 3. until 'changes' are minimal. (The log-likelihood of the model is monitored for convergence)

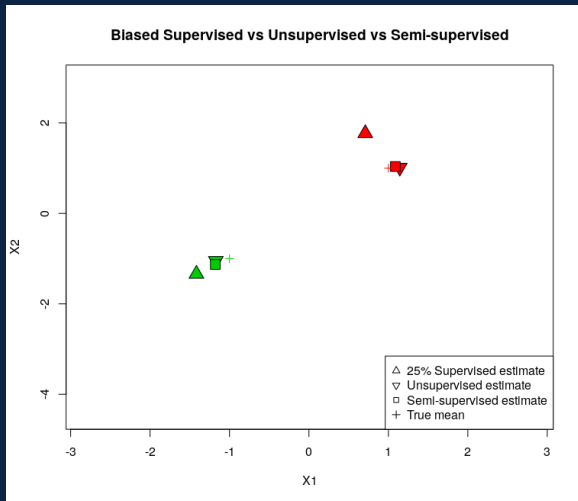
25% Supervised vs Unsupervised vs Semi-Supervised



- We can reanalyze those same simulations from before using this paradigm.



Bias Supervised vs Unsupervised vs Semi-Supervised





- ▶ More recent versions of MCLUST can run supervised classification through `MclustDA()` and semi-supervised through `MclustSSC()`.
- ▶ tEIGEN can be used semi-supervised by inputting NA's for the unlabeled observations in the “known” vector.



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